

Conclusion

Summary of Contributions I

This paper has applied the Cognitive Integrity Framework (CIF) [?] across ten critical domains, demonstrating that Goal Hijacking is not a narrow linguistic exploit but a structural corruption of autonomous decision-making. The specific contributions are:

C1: CIF-AD-OODA Integration Model. We formalized the integration of three complementary frameworks—CIF (defense mechanisms), Axiomatic Design (structural analysis) [?], and the OODA Loop (temporal dynamics) [?—into a unified analytical model for Goal Hijacking. This model enables systematic domain analysis through a standardized five-step template.

Summary of Contributions II

C2: Universal Attack Pattern Taxonomy. Through cross-domain analysis, we identified three universal attack patterns—FR Polarity Inversion, Constraint Relaxation, and Context Boundary Violation—that characterize all Goal Hijacking attacks as specific manipulations of the Axiomatic Design Matrix. FR Polarity Inversion is the most prevalent (5/10 domains), revealing that the most effective attacks *co-opt* rather than *disable* agent capabilities.

C3: CIF Mechanism Coverage Validation. We demonstrated that all five canonical CIF mechanisms appear across the ten-domain portfolio, with each mechanism serving as a primary defense in at least three domains. No domain requires mechanisms beyond the CIF vocabulary, and no single mechanism suffices alone—confirming Paper 1’s defense-in-depth architecture.

Summary of Contributions III

C4: Novel Defense Patterns. Three domains contributed genuinely novel extensions to the CIF vocabulary: *verification channel separation* (Biowarfare), *active perturbation probing* (Trade Wars), and *physics-informed invariants* (Infrastructure). These patterns generalize beyond their originating domains and represent candidate additions to the canonical CIF mechanism set.

C5: Temporal Scale Analysis. The OODA transient dynamics analysis revealed that Goal Hijacking operates across eight orders of magnitude in time scale (milliseconds for drone swarms to years for diplomatic agents), demonstrating that CIF's temporal parameters (ϵ , Δt) must be domain-calibrated but the underlying defense principles are scale-invariant.

Summary of Contributions IV

C6: Real-World Validation. Retrospective analysis of six documented AI agent security incidents (2024–2025)—including the Replit agent meltdown, GitHub Copilot RCE (CVE-2025-53773), Slack AI data exfiltration, and a \$3.2M procurement fraud—confirms that all incidents map to one of the three universal attack patterns and would have been detectable or preventable by the appropriate CIF mechanism. This provides the first empirical grounding for the CIF-AD-OODA framework in real production failures (see Supplementary Material S2).

Relationship to the Series I

This paper completes a four-part arc in the *Cognitive Security for Multiagent Operators* series:

- ▶ **Paper 1: Formal Foundations** [?] (DOI: 10.5281/zenodo.18364119) established the formal foundations: cognitive state model $\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{J}_i, \mathcal{H}_i \rangle$, trust calculus with δ^d bounded delegation, adversary taxonomy (Ω_1 – Ω_5), information-theoretic stealth–impact bounds, and five canonical defense mechanisms with composition algebra. A supplementary chapter additionally develops the eusocial-colony analogy.
- ▶ **Paper 2: Computational Validation** [?] (DOI: 10.5281/zenodo.18364128) provided computational validation: benchmark evaluation across 950 attack scenarios, ablation studies, Bayesian uncertainty quantification, and colony-scale benchmarks, with the recommended defense stack achieving 94–100% detection in parametric simulation.

Relationship to the Series II

- ▶ **Paper 3: A Qualitative Review for Practitioners [?]** (DOI: 10.5281/zenodo.18364130) translated the formal and empirical results into accessible engineering guidance: deployment guides, subagent-hardening patterns, incident-response playbooks, monitoring strategies, cost–benefit analysis, common pitfalls, case studies, and operator risk frameworks. No formal prerequisites assumed.
- ▶ **Paper 4: Applications of the Cognitive Integrity Framework** (this paper) demonstrates real-world applicability: CIF's formal mechanisms map to concrete defenses across ten high-stakes operational domains via the integrated CIF-AD-OODA model, and the cross-domain analysis yields new insights (three universal attack patterns, three novel defense extensions) not visible from any single domain, with retrospective validation against six documented 2024–2025 AI agent security incidents.

Relationship to the Series III

Together, the series establishes that cognitive integrity is not merely a theoretical concern but a *necessary engineering discipline* for deployed multiagent systems. Readers seeking derivations or proofs should consult Part 1; readers seeking empirical measurement should consult Part 2; readers looking to deploy these defenses operationally should consult Part 3; readers evaluating CIF for a specific operational sector have just read the present work.

Future Work I

Several directions emerge from this analysis:

1. **Empirical validation.** The most critical next step is controlled experimentation in at least one domain—ideally cyber-security or infrastructure, where testbed environments exist—to validate CIF defense effectiveness against real Goal Hijacking attacks with measured detection rates and false positive costs. The real-world incidents cataloged in Supplementary Material S2 provide natural experiment data for retrospective validation—particularly the Replit and procurement agent cases, where the full attack chain is documented and the hypothesized CIF defenses can be simulated against the recorded agent behavior.

Future Work II

2. **Multi-domain attacks.** Adversaries operating across domain boundaries (e.g., manipulating food security data to influence trade policy agents) represent a class of attacks that single-domain analysis cannot capture. Federated CIF architectures with cross-domain trust management are needed.
3. **CIF parameter tuning.** Systematic derivation of optimal mechanism parameters (τ , ϵ , q , Δt) for each domain, potentially through automated calibration against domain-specific attack distributions.
4. **Automated domain analysis.** The five-step domain analysis template is currently applied manually. Automation—where an AI agent characterizes its own operational context, identifies its FRs and DPs, and selects appropriate CIF mechanisms—would enable self-configuring cognitive integrity.

Future Work III

5. **Higher-class adversaries.** Extending the applied analysis to Ω_3 – Ω_5 attacks and multi-class compositions would address the scope limitation identified in `sec:limitationsdiscussion`.
5. **Tool ecosystem security.** The emergence of the Model Context Protocol (MCP) and similar agent-tool integration frameworks introduces attack surfaces—particularly tool poisoning and tool-call interception—not captured by the current Ω_2 analysis. As tool ecosystems become the primary interface between agents and their operational environments, CIF mechanisms must be extended to cover the tool integration layer explicitly.

Future Work IV

Dual-Use Considerations

We note that the CIF-AD-OODA framework, while designed as a defense methodology, also serves as an analytical tool that could assist adversaries in identifying undefended attack vectors. The universal attack pattern taxonomy (C2) and the CIF mechanism coverage matrix

(sec:mechanism_coverage) collectively identify which domains are most vulnerable

Closing Statement I

Prior to this work, the threat of Goal Hijacking was largely viewed as an issue of content moderation—preventing a chatbot from saying something inappropriate. We have demonstrated that in the domain of deployed multiagent systems, Goal Hijacking is a kinetic and existential threat. It is the ability of an adversary to rewrite the Functional Requirements of our critical infrastructure, turning our own autonomous agents against us.

Closing Statement II

By integrating Axiomatic Design principles with the Cognitive Integrity Framework and analyzing the temporal dynamics through the OODA lens, we establish both a theoretical foundation and a practical defense methodology. We move from fragile “prompt engineering” to robust “goal engineering.” We secure the OODA loop not by sanitizing the world of adversarial inputs, but by hardening the agent’s Orientation phase against the transient seduction of the hijack—through Cognitive Firewalls that filter, Belief Sandboxes that isolate, Behavioral Invariants that constrain, Drift Detectors that monitor, and Byzantine Consensus that validates. In doing so, we ensure that as our systems become faster and more autonomous, they remain unmistakably *ours*.